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Agam Baruah

Education

- 2024–Present **BS–MS (Dual Degree)**, *IISER Pune*, India, Current CGPA: 9.1/10
- Semester 1: 9.0/10, Semester 2: 9.5/10, Semester 3: 9.0/10, Semester 4: 9.0/10
 - Relevant Coursework: Introductory Quantum Physics, Mathematical Methods for Physics I & II, Classical Mechanics, Thermal & Statistical Physics, Data Analysis, Introduction to Computing, Thermodynamics
 - In Progress (Sem 5): Quantum Mechanics-I, Statistical Mechanics-I, Optics, Electronics, Electrodynamics
- 2012–2024 **Senior Secondary (10+2)**, *Kendriya Vidyalaya CUT, Tezpur University Campus*, India
- Class 10: 98.2% (CBSE)
 - Class 12: 95.6% (CBSE)
 - Subjects: Physics, Chemistry, Biology, Mathematics, English

Research Experience

Astrophysics & Data Analysis Experience

- May 2026 – **FRBs as Probes of the Cosmic Baryon Distribution**, *Summer Research Fellow, IISc*
July 2026 *Bangalore*, Supervisor: Dr. Prateek Sharma
- Compiled and analysed a sample of non-repeating FRBs from the CHIME/FRB Catalog 1, applying signal-to-noise ratio thresholds to obtain a clean cosmological sample
 - Computed DM_{cosmic} distributions and performed spatial cross-correlation of FRB positions with DESI and SDSS galaxy catalogues in 0.5° boxes, identifying a statistically significant correlation between FRB excess and galaxy overdensity
 - Studied how FRB dispersion measures constrain the large-scale distribution of baryonic matter, building on the Macquart et al. (2020) relation and Connor et al. (2025) methodology
 - Presented findings to Dr. Sharma's research group; prepared a full written project report
- Dec 2024 – **Winter School in Astronomical Calculations Using Python**, *Wilson College, Mumbai*,
Jan 2025 Grade A+
- Performed photometric data analysis on SDSS datasets using Astropy and Pandas
 - Identified variable stars from SDSS photometric data using Python-based analysis
 - Analysed light curves and interpreted observational data

Other Research Experience

- Summer 2025 **Reading Project: Quantum Dots and Applications**, *IISER Pune*, Prof. Pramod Pillai, Nano Alchemy Lab
- Conducted literature review on photocatalytic systems
 - Analysed reaction mechanisms and data interpretation
 - Prepared a technical literature review report

Other Experiences

- Jul 2025 – **Multimedia Team Member**, *GalaxyZ Space*, Internship
- Aug 2025 ○ Designed the organisation logo and created science communication graphics for the official Instagram channel
- Aug 2025 – **Design & Content Coordinator**, *Aakashganga, Astronomy Club of IISER Pune*, IISER Pune
- Aug 2026 ○ Outreach and event promotion
- March 2026 – **Member**, *ANYmation – Astrophysical Visualization Project*, IIT Bombay
- Present ○ Astrophysical visualisation using Python and Blender

Conference Participation & Workshops

- March 2025 **Second Daksha Science Workshop**, *Mumbai, India*
Invited participant in a competitive astrophysics workshop with collaborative data analysis projects

Technical Skills

- Programming Python (emcee, NumPy, SciPy, Pandas, Seaborn, Matplotlib, Astropy, Lightkurve)
- Tools Linux (Shell, Basic command-line operations), Jupyter Notebook
- Data Analysis SDSS Datasets, CHIME/FRB Catalog, DESI & SDSS Galaxy Catalogues

Honors & Achievements

- INSPIRE Scholarship – Awarded to top 0.1% of Indian STEM students by Govt. of India, Department of Science and Technology (January 2025)
- International Astronomy and Astrophysics Competition (IAAC) – Qualified participant, recognised globally for analytical skills (July 2025)
- JEE Advanced 2024: CRL Rank 20059; JEE Mains 2024: 97.8 percentile; IISER Aptitude Test (IAT) 2024: Rank 399